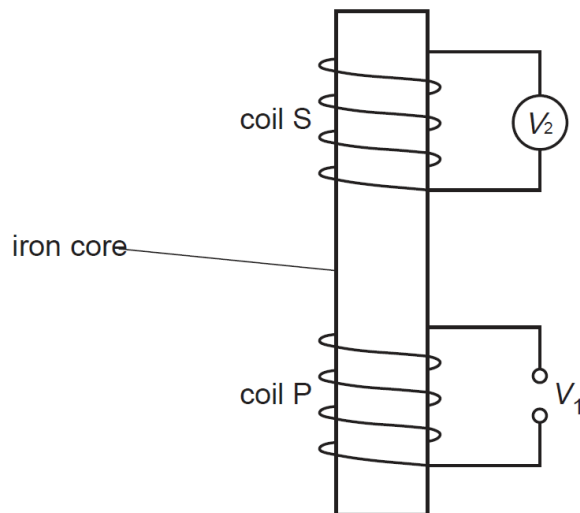
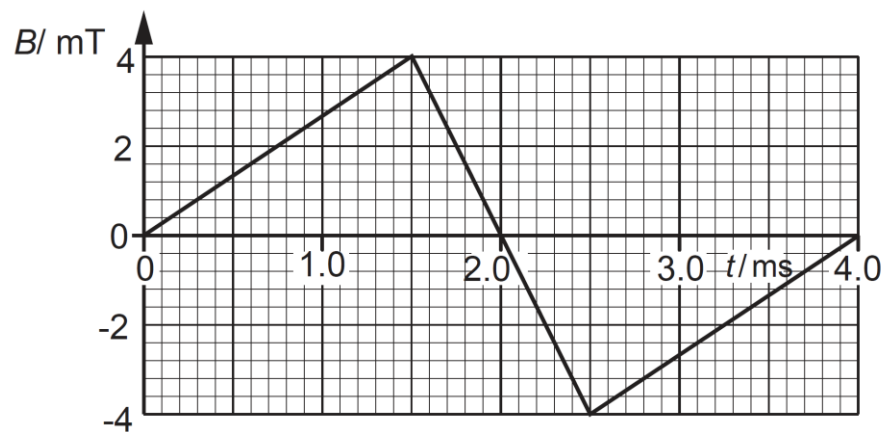


5	(a)	State Faraday's law of electromagnetic induction.	
		[2]	
		Solution: The magnitude of the electromotive force (e.m.f.) induced in a conductor is directly proportional to the rate of change of magnetic flux linkage of the conductor.	1 1
	(b)	Two coils, P and S, are wound on an iron core, as shown in Fig. 5.1.	

**Fig. 5.1**

Coil P is connected to a potential difference (p.d.) V_1 that gives rise to a magnetic field in the iron core.

Fig. 5.2 shows the variation with time t of the magnetic flux density B in the iron core.

**Fig. 5.2**

Coil S contains 250 turns of wire. Each turn of wire has a diameter of 2.5 cm. A voltmeter measures the electromotive force (e.m.f.) V_2 that is induced across coil S.

Use data from Fig. 5.2 to determine the maximum electromotive force (e.m.f.) induced in coil S.

e.m.f. = V [3]

Solution:**Max gradient of the B-t graph**

$$= \frac{dB}{dt} = \frac{(4 - (-4)) \times 10^{-3}}{(2.5 - 1.5) \times 10^{-3}} = 8 \text{ T s}^{-1}$$

Maximum induced e.m.f. in coil S

$$= |E| = \frac{dNBA}{dt} = NA \frac{dB}{dt}$$

$$= N \times \pi \times \left(\frac{D}{2}\right)^2 \times \frac{dB}{dt}_{\text{max}}$$

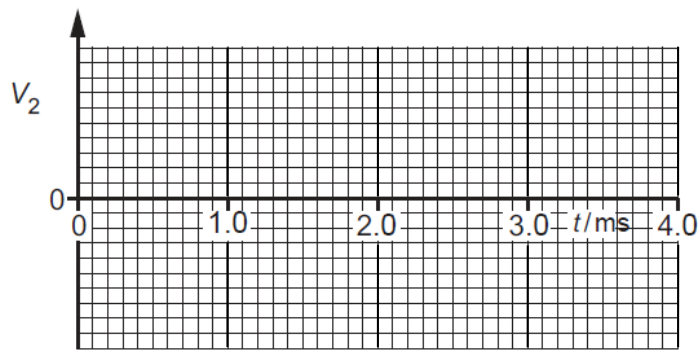
$$= 250 \times \pi \times \left(\frac{2.5 \times 10^{-2}}{2}\right)^2 \times 8$$

$$= 0.982 \text{ V}$$

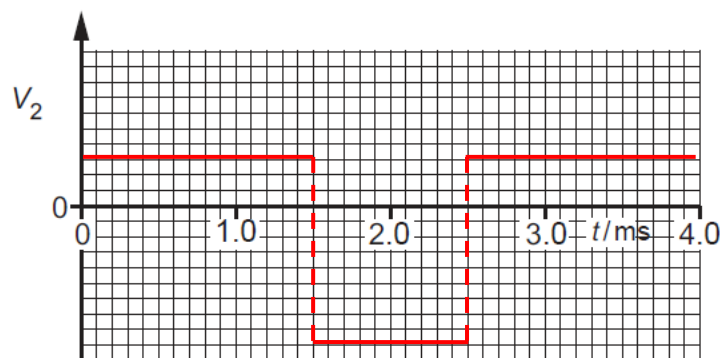
1

1

1

(c) On Fig. 5.3, sketch the variation with t of V_2 between $t = 0$ and $t = 4.0$ ms.**Fig. 5.3**

[4]

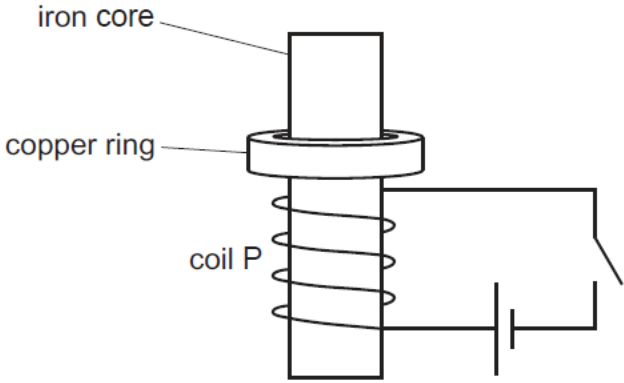
Solution:**Fig. 5.3**

V_2 stepped, all at non-zero values, between $t = 0$ and $t = 0.40$ s (horizontal line for all 3 segments) – Marker: not time based

1

V_2 shown with same non-zero magnitude up to $t = 0.15$ s and after $t = 0.25$ s but with a different magnitude between these times (Marker: either a hill or valley)

1

		V_2 shown with a magnitude between $t = 0.15$ s and $t = 0.25$ s that is three times the magnitude before $t = 0.15$ s and after $t = 0.25$ s	1
		V_2 shown with same sign up to $t = 0.15$ s and after $t = 0.25$ s, and opposite sign in between	1
	(d)	Coil S in (b) is now replaced with a copper ring that rests loosely on top of coil P. The supply to coil P is replaced with a cell and a switch that is initially open, as shown in Fig. 5.4.  Fig. 5.4	
	(i)	The switch is now closed. As it is closed, the copper ring is observed to move vertically upwards on the iron core. Explain why this happens. [4]	
		Solution: Increasing/ changing current in coil P causes increasing/ changing magnetic flux density in coil iron core links the magnetic flux density/ field from coil P to the ring Increasing/changing (magnetic) flux linkage experienced by the ring causes induced e.m.f. in ring by Faraday's law and this causes a current in ring (closed circuit) (magnetic) field due to (induced) current in ring interacts with (coil's) field to oppose the change to cause an upwards force (on ring) according to Lenz's law.	1 1 1 1
	(ii)	Suggest, with a reason, what would be the effect on the ring of repeating the procedure in (d)(i) with the terminals of the cell reversed.	
			[1]

			Solution: Both magnetic fields reverse direction so ring still moves up or current (in ring) and (coil's) field both reverse so ring still moves up or when terminals are reverse, there will still be a changing flux experienced by the ring so ring still moves up.	1
--	--	--	--	---

6	(a)	State how a line emission spectrum may be explained on the basis of the existence of
---	-----	--